



Literature Review Outline

Topic: Cybersecurity threats in IoT for the Power and Utilities sector (Smart Grids and SCADA)

(Research Methods and Professional Practice)

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1. Introduction

1.1 Main goal

Check how using IoT in smart-grid and SCADA setups shifts security risks, and spot controls that cut down real-world dangers.

1.2 Readers

Experts in research, power company security groups, rule makers for standards, and government watchers.

1.3 Goal and value

Help safe tech upgrades by linking proof of threats to practical fixes and guidelines.

1.4 Methodology for locating and selecting literature

Databases—IEEE Xplore, ACM DL, ScienceDirect, IET, NIST, ENISA. Keywords ‘smart grid security’, ‘SCADA/ICS security’, ‘IEC 62351’, ‘NIST SP 800-82’, ‘AMI cyber security’. Inclusion—2010–2025; peer-reviewed or standards; utility focus.

2. Conceptual foundations

- Layers and areas in smart-grids and SCADA; gaps between IT goals and OT needs.
- Key links and tools: IEC 60870-5-104, DNP3, Modbus/TCP; entry points; back systems for AMI.
- Guide framework: NIST SP 800-82; NISTIR 7628; IEC 62351.

3. Sector technological demands

- Limits in daily work: Keeping safe, steady timing, old gear life, mixed suppliers.
- Forces for change: Adding in solar/wind and electric cars, big growth in AMI, cloud data checks.
- Common wants: Safe data sends, far-off fixes, updates you can check.

4. Applications and threat scenarios

- Hacks on field tools, bad software loads, stolen codes.
- Jumps from substations through wrong remote links.
- AMI data leaks and tricks on power use patterns.
- Messing with time links and power state checks.

5. Analytical framework

- Link risks to key parts (tools, nets, apps, oversight).
- Score fixes by ease in active OT (split areas, safe links, watch tools, rules).

6. Findings

- Good effects from top fixes: Sharp borders, less side jumps, checks on trust points.
- Ongoing issues: Old machines, update gaps, tied to suppliers, staff shortages.

7. Strengths and limitations of existing work

- Pros: Firm rules and growing advice.
- Cons: Rare direct comparisons; thin info on costs and people factors.

8. Discrepancies

- Real use of strong codes at far points.
- How well watch systems work with low band and strict timing.
- Cloud use for quick tasks.
- Upgrade system to new one

9. Conclusions and recommendations

- Use protocol setups that match IEC 62351.
- Set strict areas and approved talks. • Roll out monitoring fit for OT.
- Map out shifts from old gear.
- Run regular test attacks with safety stops.

10. References

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Knowles, W., Prince, D., Hutchison, D., et al. (2015) 'A survey of cyber security management in industrial control systems', International Journal of Critical Infrastructure Protection, 9–10, pp. 52–80.

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NIST (2015) Guide to Industrial Control Systems (ICS) Security (SP 800-82 Rev. 2). Gaithersburg, MD: National Institute of Standards and Technology.